SUBMITTALS FOR POMPEY'S PILLAR INTERPRETIVE CENTER

☐ NO EXCEPTIONS NOTED
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☐ RETURN FOR CORRECTION
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ROBERT PECCIA & ASSOCIATES

Date; May 20, 2005

Dwain Salveson
CCM Inc.
PO Box 20996
1629 Ave D
Billings, MT 59104-0096

Section: 11395 Part 2

Equipment: Chem Free Iron Filter

Contact: Jon Rutt 406-628-6059 Fax 406-628-8149

Chemical Free Iron Filter System

Description

1. Concept
The removal of iron by the introduction of a small amount of air into a system, sufficient to start oxidation of the iron, but not sufficient to complete the oxidation prior to the water passing through the filter media. Here the media causes the pH to rise to approximately 8.5. As well, the iron is attracted by the catalytic effect of the media, thus removing all iron from the water.

   It removes: ferric (red water) iron; ferrous (dissolved, clear water) iron; bacterial iron and colloidal iron.

2. **Its limitations are:**

   Iron – virtually none

   pH – must have a minimum of 6.0 pH and, with pH below 7.0, the mag oxide becomes somewhat sacrificial. On low pH water, "M" units are recommended for higher pH correction.

   Hardness – no bearing. However, if hardness or raw water is less than one grain, use a standard "A" bed and you will get good pH correction from 6.5 to 8.5.

   Manganese – if you have no manganese, use the standard media bed (e.g. MC10A up to 1.0 mg/l manganese, use the special "M" media). Over 1 mg/l, the filter will not remove in excess of 1 mg/l. It will simply slip through.

   Hydrogen Sulfide – 2 to 3 ppm maximum is the limit

3. **Selling feature highlights include:**

   a. Unlike conventional filters, it removes bacterial iron.
   b. Has no upper limit for iron removal
   c. No chemicals (potassium, chlorine) required for regeneration.
   d. Maximum 50 gallons of water used for total backwash. Also, regenerates less often than conventional filters so equates to less than 10% of the water other types of filters use in regeneration.
   e. No maintenance required by consumer or dealers – truly automatic and maintenance-free.
   f. Unit has a short 16 minute cycle which allows consumer to have filtered water for all but 16 minutes of the 24 hour day.

4. **What materials does the filter consist of?**

   a. Hydrocharger (1" pipe size)
   b. Filter media consisting of magnesium oxide, calcium carbonate, fine filter sand and support gravel.
   c. 2500 fully automatic brass control valve
   d. Fiberglass pressure vessel
   e. PVC distribution tube
   f. Braukmann air vent assembly (on control valve) supplied in package.

5. **How do we introduce the controlled amount of air into the system?**
   The hydrocharger is installed ahead of the pressure tank and is set to aspirate air for a minimum of ⅓ of the pump cycle. We would suggest a minimum pump cycle of 60 seconds. The hydrocharger should be shown so that everyone understands how it functions and how to check for aspiration as well as setting the time it aspirates. To increase aspiration, turn the screw clockwise; to decrease time of aspiration, turn set screw counter–clockwise.

6. **How to measure a pumping rate properly:**
   Having sufficient pumping rate is critical. Also, the hydrocharger simply will not aspirate with less than 3 gpm pumping rate.

   Draw ______ ÷ cycle time ______ x 60 = pumping rate (in gallons per minute)

7. **Importance of installation of system:**

   a. Location of hydrocharger
   b. Location of pressure switch
   c. Location of filter
   d. Location and size of pressure tank

   It is imperative that the installation be such that the equipment be installed in the following order:

   1 – hydrocharger; 2 – pressure switch; 3 – pressure tank; 4 – iron filter. It is necessary to have a minimum of 6" of uninterrupted straight line on the inlet and outlet of the hydrocharger. Rapid pump cycling may occur if the pressure switch precedes the hydrocharger! A minimum cycle time of 60 seconds is recommended with adjustment of the hydrocharger to ⅓ of this cycle time. Should the cycle time be only 40 seconds, we would suggest setting the hydrocharger to aspirate approximately ½ the cycle time to ensure sufficient air to start oxidation process.

8. Low pH
When water has a pH of below 6.0, consult WaterGroup Inc. because recommendation may require using a chemical feeder (feeding soda ash) ahead of a Chemical Free Iron Filter. This is more likely to be necessary when you have a combination of lower than 6.0 pH, high iron (over 15 ppm) and manganese present.

9. It is extremely important that a sizing be based on pumping rate available but also service flow rates (continuous, peak) are observed.

10. Even though there are advantages in using an air–to–water pressure tank, captive air and bladder can, and are, most commonly used. This is why WaterGroup Inc. has developed and supplies an air vent system with each system!

11. **Importance of pH in oxidation**

a. Iron will readily oxidize when the pH of the water is 7.0 or higher.

b. Manganese will not readily oxidize unless the pH is 8.2 – 8.5. Therefore, we use a media with more magnesium oxide (raises pH) when there is manganese present in the water. This magnesium oxide is somewhat sacrificial when the pH is low or when manganese is present. The rule of thumb is:

pH of 6.0 – add 1 jar of MPH adder 3 times a year
pH of 6.5 – add 1 jar of MPH adder 2 times a year
pH of 7.0 or higher (no manganese present – never add MPH

Do not use iron and manganese type "M" unit when there is not manganese present in the water because you may raise the pH too high unless you have lower than 7.0 pH.

Where you are using an "M" model when there is manganese present, add MPH adder twice a year for every full measurement below 8.5 the raw water's pH is.

MPH can be added with a special ML–1 media loader which is available as an option.

12. **The 2500 control and how it functions:**

a. Why we use the 2500 versus the 5600 (adjustable cycle times).
b. Pin settings
c. Explain time of day settings, time regeneration, frequency of regeneration, program wheel and pin settings.

13. Tidbits

The unit should be backwashed right after installation so that the media bed can be totally mixed up to prevent it from cementing. It is also important to backwash the unit after MPH has been added. It is a fact that this filter, unlike conventional types, filters better when the media becomes somewhat dirty or iron–fouled. That is why we suggest following the frequency of regeneration chart in the manual. It can be detrimental to backwash a system too often. It is also important that people are aware that the first day or two after installation, some iron will slip through the system. It will only totally remove iron after approximately 2 days of service (upon installation). If you were to look at the backwash water at the end of the time allowed, it would not be clear. It is not supposed to be. Again, the filter works better when the media is slightly coated with iron. The hydrocharger can be installed vertically or horizontally, but you must be aware of the proper direction of flow of water indicated by the arrow and the 6" of uninterrupted line on the inlet and outlet.

We want our dealers to have confidence selling this product, the peace of mind knowing the products can and will do the things outlined when sold. By knowing the process and learning proper installation, we feel the dealers will have the confidence and peace of mind needed.

We feel it is extremely important to contact the manufacturer (WaterGroup Inc.) when water analysis shows extreme conditions, high iron, low pH, high total dissolved solids, etc.

SUMMARY OF EQUIPMENT

MODEL ACF 24 - 1.5FTT0-CC AUTOMATIC CHEMICAL FREE IRON FILTER

Filter Tanks
Number 3
Diameter.......... 24”
Height of Tank.......... 72”
Design Working Pressure.......... 100 psig

Filter Media
Type .... Mixed Media
consisting of ; Calcium Carbonate, Magnesium Oxide, Fine Filter Sand

Support Bed .... Graded Quartz

Distribution System.......... Hub Lateral (PVC Riser)

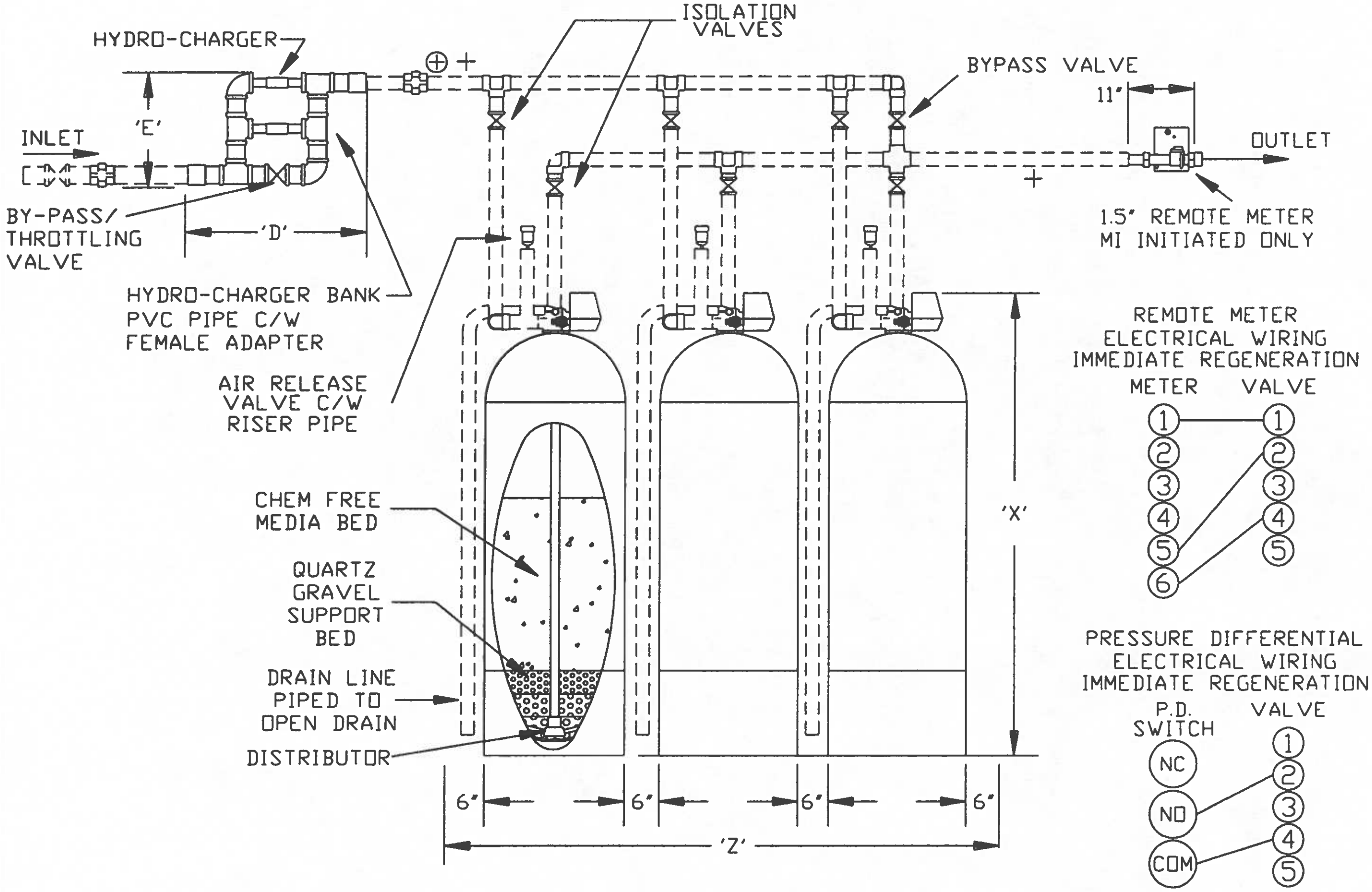
Main Piping
Size .... 1.5”
Material.......... by others
Connection .... by others

Control Valves
Size .... 1.5”
Type .... Fleck 2850 Filter
.... Top Mounted Valve

Air Induction Assembly
Number of Injectors.......... ~~1 - 1” Hydrocharger~~ 1 - 1½" Mazzei
Balancing Valve Type .... ~~PVC Ball Valve~~ Brass Globe Valve
Balancing Valve Size .... ~~1.5”~~ 2.0"

Backwash Initiation Method .... Calendar Clock

Flow Rates
Normal Service/Unit.......... 9 usgpm
Maximum Service/Unit.......... 25 usgpm
Backwash/Unit .... 30 usgpm



NOTES
1. FIBERGLASS TANKS MAX. WORKING PRESSURE IS 150 PSI AND 120° F. SYSTEM PRESSURE 100 PSI MAX. TANKS ARE NSF APPROVED.
2. ITEMS SHOWN BY BROKEN LINES ARE SUPPLIED BY OTHERS.
3. REQUIRES 120 VOLT, 60 Hz FOR ELECTRICAL POWER.
4. ALL TANKS C/W 4" TOP & BOTTOM HOLE.
5. TOP MOUNT 1.5" VALVE IS A FLECK 2850.
6. NO-UNTREATED WATER BYPASS PROVIDED BY THE NO-UNTREATED WATER BYPASS PISTON.
7. ✳ INDICATES RECOMMENDED LOCATION OF PRESSURE GAUGES AND SAMPLE COCKS.(NOT INCLUDED)
⊕ INDICATES RECOMMENDED VACUUM BREAKER LOCATION IF REQUIRED. (NOT INCLUDED)
8. MEDIA BED CONSISTS OF CALCIUM CARBONATE, MAGNESIUM OXIDE, AND FINE FILTER SAND ON A GRADED QUARTZ SUPPORT BED.
9. INITIATION TYPE IS DETERMINED BY THE LAST TWO DIGITS IN THE MODEL NUMBER "__". CC FOR CALENDAR CLOCK, MI FOR METER INITIATED, PD FOR PRESSURE DIFFERENTIAL. + INDICATES RECOMMENDED LOCATION OF PD TUBING CONNECTIONS.
10 METER INITIATED FILTER USES 1.5" METER. RECOMMENDED FLOW RANGE IS 1.5 TO 75 USGPM.
11 NORMAL SERVICE FLOWRATES BASED ON APPROX. 3 USGPM/SQ. FT.
12 BACKWASH FLOWRATES BASED ON APPROX. 10 USGPM/SQ. FT.
13 SYSTEM MUST BE DESIGNED SO THE HYDRO-CHARGER BANK DRAWS AIR PROPERLY.
14 # REFER TO MODEL NUMBER. LEAVE SPACE BLANK FOR STANDARD FILTER. PUT "M" IN FOR LOW pH OR HIGH IRON FILTERS.
15 FOR PROPER AIR RELEASE, USE A RISER PIPE BELOW THE AIR RELEASE VALVE.
16 BOTH UNITS IN SERVICE. INTERLOCKED SERIES REGENERATION. VALVES WIRED TO ALLOW THE SYSTEM TO BE EXPANDED.
17 ELECTRONIC TIMER ALSO AVAILABLE __________

U.S. FILTER

AUTOMATIC CHEMICAL FREE FIBERGLASS TRIPLEX FILTER

PROJECT NO.

DRAWING NO. CF15FTTO REV. 0

MODEL NUMBER (TRIPLEX)	TOTAL RECOMMENDED FLOWRATES			PIPE	REC. DRAIN	TANK SIZE	RECOMMENDED MIN. DIMENSIONS					TOTAL WEIGHT	
	SERVICE	PEAK	BACK-WASH	SIZE	SIZE		WIDTH (Z)	DEPTH (Y)	ACTUAL HEIGHT(X)	WIDTH (D)	HEIGHT (E)	SHIPPING	OPERATING
	USGPM	USGPM	USGPM	INCHES	INCHES	INCHES	INCHES	INCHES	INCHES	INCHES	INCHES	LBS.	LBS.
ACF 21-1.5#FTTO-__	21	36	20	1.5	1.5	21x69	87	33	85	38.5	10.5	2610	4395
ACF 24-1.5#FTTO-__	27	45	30	1.5	1.5	24x72	96	36	88	38.5	10.5	4170	6690

Problem Water Application

	Water Softener	*Compen-sation*	*Chem Free (CF)*	*Chem Free (CF-M)*	*Potassium Filter (IF)*	*Comp. Numbers*	*Notes*
Iron	1.5 ppm	x 1	35 ppm	35 ppm	7 ppm	x 1	
Manganese	0.75 mg/l	x 2 – total must be 1.5	0	1.5 ppm	5 ppm	x 2	Twice the oxygen demand of iron
H_2S	0–must be treated first		3 ppm	3 ppm	3 ppm	x 3 10 ppm or less	Chem free not good on just H_2S
pH	N/A		7 or higher	6.5 – 6.9	must be 7.0 or higher (7.2 ideal)	10 ppm or less	
Iron Bacteria	No		Yes	Yes	No		Causes service problem on units with injectors
Tannins	N/A		0.5 ppm	0.5 ppm	N/A		Must see color before being concerned
Turbidity	None		Careful – think about backwashing too often		Careful – cost of pot.perm. vs turbidity		Chem free works best dirty
Pressure (psi)	15 – 120 psi		15 – 50 maximum		15 – 120 psi		Chem free doesn't draw air if pressure settings too high

For Chem Frees:

1. Cycle Time – must be a minimum of 30 seconds
2. Water to Air Pressure Tank – whenever H_2S is present or the compensated total is higher than 10, these are recommended.

WATER SPECIALTIES
A Division of Rutts Inc.

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ROBERT PECCIA & ASSOCIATES

May 22, 2005

Dwain Salveson
CCM Inc.
PO Box 20996
1629 Ave D
Billings, MT 59104-0096

RE: Submittals for the Pompey's Pillar Interpretive Center.

I have discussed the water quality with your engineer and the submittals reflect the course of treatment decided. Attached are the submittals for the potable water pump, water treatment and related components for the pump house.

Our qualifications include the following projects:
Public Water Supply systems:

Salish Kootenai Housing Authority-Pablo, MT
SKHA Attn: Bob Gordon
PO Box 38
Pablo, MT 59855
406-675-4491
We built 3 water plants utililizing chemical feed and iron removal.
Oak Ridge Subdivision-Billings, MT

Aquanet/Dan Wells
PO Box 80445
Billings, MT 59108
406-656-1301
We built the water plant using chemical feed and pumping capable of 300 gpm.

Saddleback Subdivision-Laurel, MT
Sam Picard/Picard Development
373 Delta Circle
Billings, MT 59102
406-656-2333
We built the water plant utilizing storage, chemical feed, reverse osmosis, pumps and controls for a 100 home subdivision.

I am the General Manager of Water Specialties and Service Manager. I am in my 32nd year of working in water treatment. I am certified by the Water Quality Association as a Certified Water Specialist with endorsements in Reverse Osmosis, Deionization, Filtration, and Ultra Filtration. I attended Culligan Commercial-Industrial Systems Master Service School and graduated the top of all the classes as a Master Service Technician.

603 West Main, Laurel, MT 59044 • (406) 628-6059 • Fax (406) 628-8149

Process:

The water will be pumped from the well to the pump house. The pump will be controlled by the level controls in the cistern located under the pump house. The water will flow through a water meter and tee off a flush line. A sample port will allow for raw water sampling. An air eductor will increase the dissolved oxygen content in the raw water before proceeding to 2 40 gallon utility contact tanks. This will vent off the excess air before the water passes through the Triplex Chem Free Iron Filters. After the iron filters a minimal amount of chlorine (.5 mg/l to .7 mg/l) will be added to the water as it is discharged to the fire tank/cistern.

The backwash of the iron filters will be controlled by a time clock that will allow backwash at night with one tank backwashing every 4 days. The back wash rate is 30 gallons per minute for 7 minutes. This will be approximately 4% of throughput at the minimum usage of 1300 gallons per day. The iron filters will be backwashed by treated water. The well pump will be locked out and treated water (pre-softening) will be used to backwash the filters.

The treated water will be pumped out of the fire tank/cistern with a constant pressure variable speed pumping system, pass through a water meter, tee off to a flush line, tee off to backwash the iron filters, and pass through the water softener before heading off to the other buildings. A sample port will be provided before the softener and after the softener for quality control and sampling. The water sink in the pump house will be connected before the water leaves the building.

Components:

Air Injection will be accomplished with Mazzei model 1583 injector. The injector will flow 25.2 gpm and draw plenty of air to oxidize the iron and manganese properly. The injector will be installed in a bypass assembly. The bypass will allow us to fine tune the bypass flow to maintain flow and pressure differential required for proper air suction.

The 2 HP-9 contact tanks will allow for any excess air to vent off before the iron filters.

The Iron Filter is a Model ACF 24-1.5FTTO-CC Automatic Chemical Free Iron Filter. This unit in conjunction with the air injection will remove all the iron and manganese.

A Stenner SVP pump will be used for chemical injection with a 65 gallon solution tank.

The potable water pump is a Goulds 33GS20 with a 3 hp motor and a constant pressure controller. The pump is rated at 35 gallons per minute at 75 psi to compensate for maximum flow through the softener. Factory test certification will delay pump delivery by 4 to 5 weeks. A submersible pump operated with a variable speed controller would require a small pressure tank and transducer installed before the treatment equipment. The surge relief would not be required with a constant pressure system because of the smooth ramp up built into the controller. I would install a pressure relief set at 75 psi to blow off if overpressure occurs. This system also provides protection for low water and over pumping. The Aquavar II controller mounted with a disconnect will meet all the requirements for a control panel for the Potable Water Pump. I have attached the submittals for this unit. A sleeve would be installed over the pump to force water flow past the motor for cooling. I will submerge the pump to the bottom of the tank and install a 4 inch sleeve as per the design in the Franklin Manual. No standards are available for NPSH in a submersible pump because of the low suction ability and the small impeller eye.

I am recommending an Ecodyne S91B Duplex Alternating Water Softener. This unit will flow 36 gpm at 16 psi delta. I have attached the submittals for this unit. The VIP controller will allow for regeneration at any time. The VIP controller allows the system to look back over the last seven days and regenerate at the most convenient time without exhausting the units.

The water softener will regenerate every 3100 gallons. The regeneration water will be 105 gallons. Based on maximum flow of 6000 gallons per day, the total water required to backwash the iron filters and regenerate the softener is calculated to be 1122 gallons per 4 days. This is the frequency of the iron filter backwash. The iron filters will use 310 gallons every 4 days and the softener will regenerate 7.5 times in 4 days.

I can be contacted at 406-628-6059.

Sincerely,

Jon Rutt

Jon Rutt